

## ELEC4631 Solutions Tutorial 6

1. (a)  $c_Q(\lambda) = \lambda^2 - 6\lambda + 8$ .  $V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$  so  $V^T Q V = \text{diag}(4, 2)$  (columns of  $V$  need to be normalised for this!).

$$Q^{1/2} = V \text{diag}(2, \sqrt{2}) V^T = \begin{bmatrix} (2 + \sqrt{2})/2 & (2 - \sqrt{2})/2 \\ (2 - \sqrt{2})/2 & (2 + \sqrt{2})/2 \end{bmatrix}.$$

- (b)  $c_Q(\lambda) = (\lambda - 3)(\lambda - 4)(\lambda - 1)$ .  $V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$  so  $V^T Q V = \text{diag}(3, 4, 1)$  (again, columns of  $V$  need to be normalised for this!).

$$Q^{1/2} = V \text{diag}(\sqrt{3}, 2, 1) V^T = \begin{bmatrix} \sqrt{3}/2 + 5/6 & 1/3 & -\sqrt{3}/2 + 5/6 \\ 1/3 & 4/3 & 1/3 \\ -\sqrt{3}/2 + 5/6 & 1/3 & \sqrt{3}/2 + 5/6 \end{bmatrix}.$$

2. (a) Since  $A = 1$ ,  $B = 1$ ,  $Q = 1/2$ ,  $R = r/2$ , and  $\Gamma = 0$ , the RDE is:

$$-\dot{P}(t) = 2P(t) - 2r^{-1}P(t)^2 + 1/2,$$

with terminal condition  $P(t_f) = 0$ . The optimal control law is  $u_{\text{opt}}(t) = -2r^{-1}P(t)x(t)$ . The optimal (minimum) performance index is  $J_{t_f, \text{opt}}(x_0) = P(0)x_0^2$ .

- (b) CARE is:

$$2P - 2r^{-1}P^2 + 1/2 = 0.$$

This is a quadratic equation in  $P$ . Solving gives us two possible solutions  $P = \frac{r}{2} \pm \frac{r}{2} \sqrt{1 + \frac{1}{r}}$ . We must choose the positive solution. This is  $P = \frac{r}{2} + \frac{r}{2} \sqrt{1 + \frac{1}{r}}$ . The optimal control law is  $u_{\text{opt}}(t) = -2r^{-1}P x(t) = -\left(1 + \sqrt{1 + \frac{1}{r}}\right) x(t)$  and the optimal (minimum) performance index is  $J_{\infty, \text{opt}}(x_0) = \left(\frac{r}{2} + \frac{r}{2} \sqrt{1 + \frac{1}{r}}\right) x_0^2$ .

Note that this simple one state system gives another simple demonstration of what happens when you vary the positive parameter  $r$  which penalises the control effort. If  $r$  is sufficiently large we see that (approximately)  $J_\infty(x_0) \approx rx_0^2$ , so  $J_\infty(x_0)$  will be large. Intuitively, the reason for this is that for large  $r$  you are putting a high penalty on your control law, hence restricting what can be achieved by the controller. This leads to a large optimal value for the performance index. Conversely, for sufficiently small but non-zero  $r$  you can see that (again, approximately)  $J_\infty(x_0) \approx \frac{1}{2}\sqrt{r}(\sqrt{r} + 1)x_0^2$ , leading to a small optimal performance index. Again, this agrees with intuition, making  $r$  small means placing less restriction on the control law (e.g., it can be large), giving it more freedom to achieve the control objective.

3. (a)  $Q = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \text{diag}(0, 5) \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$ . So,

$$Q^{1/2} = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \text{diag}(0, \sqrt{5}) \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Then

$$\mathcal{O}(Q^{1/2}, A) = \begin{bmatrix} Q^{1/2} \\ Q^{1/2}A \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 5 & 10 \\ 10 & 20 \end{bmatrix}.$$

Thus,  $\text{rank}(\mathcal{O})(Q^{1/2}, A) = 1 < 2$  (not full rank), so  $(Q^{1/2}, A)$  is not an observable pair.

(b)  $Q^{1/2}$  has already been computed in sub-problem (a). Computing  $\mathcal{O}$  gives:

$$\mathcal{O}(Q^{1/2}, A) = \begin{bmatrix} Q^{1/2} \\ Q^{1/2}A \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 4 & 1 \\ 8 & 2 \end{bmatrix}.$$

Thus,  $\text{rank}(\mathcal{O})(Q^{1/2}, A) = 2$  (is full rank), and  $(Q^{1/2}, A)$  is an observable pair.

4. (a) Let

$$P(t) = \begin{bmatrix} p_1(t) & p_2(t) \\ p_2(t) & p_3(t) \end{bmatrix}.$$

Then the RDE is:

$$\begin{aligned}
& - \begin{bmatrix} \dot{p}_1(t) & \dot{p}_2(t) \\ \dot{p}_2(t) & \dot{p}_3(t) \end{bmatrix} \\
& = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} p_1(t) & p_2(t) \\ p_2(t) & p_3(t) \end{bmatrix} + \begin{bmatrix} p_1(t) & p_2(t) \\ p_2(t) & p_3(t) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\
& \quad - \begin{bmatrix} p_1(t) & p_2(t) \\ p_2(t) & p_3(t) \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} p_1(t) & p_2(t) \\ p_2(t) & p_3(t) \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \\
& = \begin{bmatrix} 2p_1(t) - (p_1(t) + p_2(t))^2 + 1 & -(p_1(t) + p_2(t))(p_2(t) + p_3(t)) \\ -(p_1(t) + p_2(t))(p_2(t) + p_3(t)) & -2p_3(t) - (p_2(t) + p_3(t))^2 + 1 \end{bmatrix}.
\end{aligned}$$

with the terminal condition:

$$\begin{bmatrix} p_1(t_f) & p_2(t_f) \\ p_2(t_f) & p_3(t_f) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

So  $p_1(t)$ ,  $p_2(t)$  and  $p_3(t)$  satisfy the coupled differential equations:

$$\begin{aligned}
-\dot{p}_1(t) &= 2p_1(t) - (p_1(t) + p_2(t))^2 + 1, \\
-\dot{p}_2(t) &= -(p_1(t) + p_2(t))(p_2(t) + p_3(t)), \\
-\dot{p}_3(t) &= -2p_3(t) - (p_2(t) + p_3(t))^2 + 1,
\end{aligned}$$

with terminal conditions  $p_1(t_f) = 0$ ,  $p_2(t_f) = 0$ , and  $p_3(t_f) = 1$ . The optimal control law is  $u_{\text{opt}}(t) = -[1 \ 1]P(t)x(t) = -[p_1(t) + p_2(t) \ p_2(t) + p_3(t)]x(t)$ , and the optimal performance index is  $J_{\text{opt}}(x_0) = x_0^T P(0)x_0$ .

- (b) The controllability matrix is  $\mathcal{O} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ , it is invertible so the system is controllable. For the pair  $(Q^{1/2}, A)$  we have that

$$\mathcal{O}(Q^{1/2}, A) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix},$$

which is full rank, so the pair  $(Q^{1/2}, A)$  is an observable pair. Since the system is controllable and  $(Q^{1/2}, A)$  is observable, the infinite horizon LQR problem has a solution.

The CARE is

$$\begin{bmatrix} 2p_1 - (p_1 + p_2)^2 + 1 & -(p_1 + p_2)(p_2 + p_3) \\ -(p_1 + p_2)(p_2 + p_3) & -2p_3 - (p_2 + p_3)^2 + 1 \end{bmatrix} = 0.$$

Giving three coupled algebraic equations:

$$\begin{aligned} 2p_1 - (p_1 + p_2)^2 + 1 &= 0, \\ -(p_1 + p_2)(p_2 + p_3) &= 0, \\ -2p_3 - (p_2 + p_3)^2 + 1 &= 0. \end{aligned}$$

Keep in mind that we need  $P = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix} > 0$  ( $P$  is positive definite). In particular, we must have  $p_1 > 0$  and  $p_3 > 0$  (**remark:** the other condition is  $p_1 - p_2^2 p_3^{-1} > 0$  or, equivalently,  $p_3 - p_2^2 p_1^{-1} > 0$ . This is easy to see. Note that  $c_P(\lambda) = \lambda^2 - (p_1 + p_3)\lambda + p_1 p_3 - p_2^2$ . So, if  $p_1 > 0$  and  $p_3 > 0$  all the roots of  $c_P(\lambda)$  will be positive if and only if  $p_1 p_3 - p_2^2 > 0$ , which is equivalent to either  $p_1 - p_2^2 p_3^{-1} > 0$  or  $p_3 - p_2^2 p_1^{-1} > 0$ . Another way to see this will be discussed in Lecture 9 on LMIs). From the second equation, it's either  $p_1 + p_2 = 0$  or/and  $p_2 + p_3 = 0$ . The first will lead to  $p_1 < 0$  from the first equation, so this has to be excluded. Thus,  $p_2 + p_3 = 0 \rightarrow p_2 = -p_3$ , and from the third equation we get  $p_3 = 1/2$ . So, this means that  $p_2 = -1/2$ . Substituting the value of  $p_2$  into the first equation gives the quadratic equation  $-p_1^2 + 3p_1 + 3/4 = 0$ . There are two solutions to this equation  $p_1 = \frac{3 \pm \sqrt{12}}{2}$ . The only positive one is  $p_1 = \frac{3 + \sqrt{12}}{2}$ . So, we have now found the values for  $p_1$ ,  $p_2$ , and  $p_3$ . Also, these values satisfy  $p_1 - p_2^2 p_3^{-1} > 0$ , so we verify that indeed  $P > 0$  as it should be.

The optimal control law is

$$u_{\text{opt}}(t) = - \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{3 + \sqrt{12}}{2} & -1/2 \\ -1/2 & 1/2 \end{bmatrix} x(t) = \begin{bmatrix} 1/2 - \frac{3 + \sqrt{12}}{2} & 0 \end{bmatrix} x(t).$$

The optimal performance index is  $J_{\infty, \text{opt}}(x_0) = x_0^T \begin{bmatrix} \frac{3 + \sqrt{12}}{2} & -1/2 \\ -1/2 & 1/2 \end{bmatrix} x_0$ .